Rhythm Karaoke: a tapping game to control vocal resynthesis of songs for dexterity rehabilitation

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Fig. 1. A user playing Rhythm Karaoke on the Dextrain Manipulandum (DM) with the right hand for two stimuli in our pilot study. (Left) The user controls the syllabic timing in the vocal resynthesis of a 16-syllable song segment by pressing a button on the DM. (Right) the user controls the onset of 16 beeps with the same DM button. The left hand presses a key on the keyboard to listen to a reference recording. Visual feedback for the user's position in each sequence is displayed on the screen.

We present a finger-tapping musical game that enables the vocal line of any song to be extracted and replayed in real-time with a new syllabic rhythm specified by the user's tapping. This game was conceived for the rehabilitation of fine manual dexterity for stroke patients and the elderly, where rhythm control is a core component.

Data of 10 healthy subjects was gathered in a pilot study, where subjects imitated rhythmic sequences of different lengths, with the lyrics of a song or beeps. Sequence length was found to have a significant effect on accuracy for both conditions, but no significant difference was found between performances across song or beep conditions. However, subjects reported higher satisfaction and lower perceived difficulty for the song over beeps. These results suggest that the inclusion of speech and music can make rhythmic exercises more engaging and motivating for users.

CCS Concepts: • **Human-centered computing** → *Auditory feedback*; *User studies*.

Additional Key Words and Phrases: tapping, dexterity, vocal synthesis, audio feedback, rhythm

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1 INTRODUCTION

The use of auditory rhythms has shown positive results in the rehabilitation of motor skills for patients of neurological disorders [20]. For gross motor skills such as gait and upper arm movement, a number of studies have shown beat

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cueing, with a metronome or music, to be beneficial for stroke and Parkinson's patients [14, 19, 21, 24]. For fine motor control, the incorporation of music has been shown to improve both task performance and motivation for stroke patients compared to conventional training methods [6].

The ability to precisely produce rhythmic movements is considered a key dimension of manual dexterity [23]. Fine motor control and precise rhythmic movements is also present in speech production [26], and our research considers how performance in the rhythmic dimension of manual dexterity training exercises can be enhanced with the incorporation of speech content.

Speech features perceptible rhythmic patterns on a syllabic level [16], and untrained subjects exhibit the ability to entrain their tapping to the syllabic rhythm of spoken phrases [8]. Cortical activity seems to show privileged perception for rhythms of a person's native language. Electroencephalography (EEG) of subjects listening to phrases in their native language displayed stronger temporal binding than for foreign phrases [15]. Speech in the context of music has also been shown to affect the brain differently than everyday spoken speech, as demonstrated by [22].

This paper presents a system, Rhythm Karaoke, that enables the vocal part of a song to be played back in real-time with a new syllabic rhythm controlled by the user's tapping. Section 2 describes prior work on tapping studies, speech rhythm, and real-time vocal synthesis. We then describe the design and implementation of Rhythm Karaoke as well as a semi-automatic pipeline for any vocal solo song to be adapted for its use (Section 3).

A pilot study is described in Section 4, where 10 healthy subjects used Rhythm Karaoke to imitate rhythmic sequences of various lengths with either the lyrics of a well-known song (speech) or beeps (non-speech). Despite the same rhythms being presented in the two conditions, we hypothesized that song lyrics will result in better overall performance because they provide temporal and contextual cues that the beeps do not. These cues include the familiar rhythmic clusters of native speech and the implied temporal grid of the song. We also expect the speech condition to yield a steeper learning curve across repetitions because the additional semantic meaning of the words can help encode the sequence into memory, especially as sequences increase in length. Results of the study are presented and discussed in Sections 5 and 6.

2 RELATED WORK

Rhythm Karaoke exploits sensorimotor synchronization (auditory-motor synchronization) for a speech and musical rhythmic task. A number of studies on sensorimotor synchronization in the context of tapping have shown that most subjects even without special training are able to reproduce rhythmic patterns in the range of 100 to 1800ms using finger tapping [17, 18]. However, these studies describe limits in accuracy, and a tendency for gestures to anticipate sequence tones by 10s of milliseconds. This tendency is decreased in musically trained subjects compared to their untrained counterparts [2].

In terms of rhythm in speech, two main rhythmic scales can be considered: intonational rhythm and syllabic rhythm [26]. Intonational rhythm corresponds to intonation contours or musical phrases, delineated by pauses. It is built on a smaller rhythmic layer, the syllabic rhythm, which is addressed by Rhythm Karaoke. In terms of perception, the syllabic rhythm can be described using the concept of the P-center (Perceptual Center), one per syllable, located near the vowel onset [7].

Rhythm Karaoke is built on enabling technology in performative vocal synthesis (PVS), or real-time, gesture controlled vocal synthesis [5, 10]. PVS systems were originally developed in the context of new musical instruments, and experiments were conducted on the most accurate methods for controlling syllabic rhythm [4, 5, 28]. Research in PVS found that reproducing syllabic rhythm to accurately emulate singing requires two movements, or a biphasic movement,

for syllabic control. This biphasic motion corresponds to the opening and closing of the vocal tract somewhere along its path [11].

Recent systems for performative vocal synthesis feature a syllabic control mode based on rhythmic control points [4, 9, 10]. Typically, two points are associated with each syllable, one (VP) for the vocalic nucleus, which is related to the P-center, and one (CP) for the inter-syllabic consonant clusters [11]. This bi-phasic movement can be mapped to the press and release of a button (on/off), where the raising and lowering of the finger is also biphasic.

The present research does not aim to understand the complex processes involved in tapping, such as anticipation or individual strategies. Nor does it seek to determine the most accurate method for controlling resynthesized speech rhythm. We take inspiration from prior studies and existing systems for our goal of creating an engaging rhythmic game involving music and speech for the purpose of dexterity rehabilitation.

3 DESIGN & IMPLEMENTATION

Rhythm Karaoke allows the user to playback the vocal line of any song by tapping to the rhythm of its syllables. Users can listen to a reference recording to practice imitating the original rhythm of the song, or play the song with their own interpretation.

Rhythm Karaoke was designed as a manual dexterity rehabilitation exercise for the hardware interface Dextrain Manipulandum DM [23]. The user interface runs in Unity, and sound is generated by the Voks vocal synthesizer [10], using OSC (Open Sound Control) to communicate between the programs.

Fig. 2. (Left) Closeup of the Dextrain Manipulandum (DM). (Right) System Diagram of Rhythm Karaoke.

3.1 Hardware Interface

The Dextrain Manipulandum (DM), shown in figure 2, is a device that accurately measures different dimensions of manual dexterity such as force control, rhythmic precision, and ability to perform individuated finger movements [23]. A precise, quantitative, multi-dimensional profile of dexterity can help clinicians identify problems that conventional measures of manual dexterity are not able to detect [25], and allow follow-up of dexterity changes over time. In addition to being a measurement tool, the DM can also serve as a training tool for patients to practice the impaired dexterity components. Rhythm Karaoke is part of a broader project aimed to add a musical and playful dimension to rehabilitation exercises on the DM.

3.2 User Interface

The user interface for Rhythm Karaoke was created in Unity, the software platform for the DM. In our current prototype, the lyrics of a song segment to be played are first displayed in black on the screen. Each time the user presses the index finger button on the DM, an OSC message is sent to Voks triggering the next syllable to be pronounced, and the letters corresponding to the pronounced syllable turn red. At the end of the song segment, the song automatically resets to the beginning, and the lyrics turn black again. The user can also press the "R" key on the keyboard to listen to a reference recording of the song segment or press the "N" key to reset the playback to the beginning of the song.

3.3 Vocal Synthesis

Voks is a vocal synthesizer programmed in MaxMSP that allows a pre-recorded vocal sample to be played back with new pitch, vocal effort, and syllabic rhythm based on real-time user input [9, 10]. Rhythm Karaoke gives users the control of syllabic rhythm, retaining the pitch (melody) and vocal effort (volume) changes of the original recording. The binary rhythm mode of Voks is used, which relies on the biphasic model described in Section 2. Binary mode requires a syllabic labeling file that specifies a sequence of time stamps of vocalic nucleii (VP) and consonant clusters (CP). An example labeling file is shown in 4. For biphasic control in Rhythm Karaoke, the VP are mapped to button presses of the DM and CP are mapped to button releases.

In addition to biphasic control, Rhythm Karaoke also supports a simpler mode with a single tap per syllable. We call this mode monophasic mode. Pressing the button in this mode triggers advancement to the next VP, with any in-between CP sent immediately before the VP. The synthesis stays on the VP, prolonging the vowel sound, until the next button press, even if the button releases in between. The articulation speed parameter in Voks is set to a high value, so that consonants are pronounced with much shorter duration than in typical speech. This adjustment is needed to minimize the latency between the button press and the VP onset due to any preceding consonants.

3.4 Data Preparation

Previous work with Voks required labeling files to be created manually [10, 28]. We devised a pipeline to semi-automatically prepare any vocal song in .wav format for use with Rhythm Karaoke, including automations to the process to create Voks labeling files (Figure 3). First, the vocal line of the song is separated from the instrumental accompaniment using Vocal Remover, a free, online AI-based source separation application [1]. Next, phoneme segmentation is performed on the extracted vocal line using Montréal Forced Aligner [13], resulting in a Praat TextGrid file that indicates the start and end timestamps of each phoneme produced by the voice [3]. The automatically generated TextGrid is then manually inspected in Praat with its corresponding .wav, and necessary corrections are made to the segmentation (Figure 4). Finally, a Python script converts the TextGrid to the labeling format required by Voks based on the algorithm specified in [9]. Currently, which parts of the lyrics are highlighted for each syllable are manually specified in the labeling file, but this can be automatically generated from the TextGrid file based on pronunciation rules.

4 PILOT STUDY

A pilot study on Rhythm Karaoke was conducted with 10 healthy native Francophone subjects (9 male, 1 female, ages 20-30, mean age: 23) to investigate the following research questions:

(1) Is there a significant difference in subjects performance (overall performance and learning curve) in a rhythm imitation task when the rhythm is carried by musical speech (i.e. song lyrics) versus non-lexical content (i.e. a beep)?
(2) Are there significant differences in performance based on the length of the target rhythm?
(3) Does the use of song lyrics versus beep influence subjects' enjoyment of the task?

4.1 Stimuli

To address question one, a non-lexical version of Rhythm Karaoke was implemented using a beep as the carrier sound. For each syllabic condition of the song, a beep reference recording was generated, with beeps triggered based on the VP labels of each syllable in the song. Voks controlled by OSC messages was also used for the beep condition to ensure equal built-in latency between the two conditions. The beep condition had a visual feedback with a row of circles corresponding to the number of beeps in the target phrase. Every time the user pressed the button on the DM to trigger the next beep, the visual feedback updates to highlight the next circle.

A French song, "Il en faut peu pour être heureux[1]," was selected for the task because of its prominent rhythm and its familiarity for the age group of our subjects. The first 8 bars of the song was used, with 26 total syllables. For consistency with the beep condition, syllabic control used monophasic mode (see Section 3.3) with an articulation speed of 4, where consonant clusters were produced at 4 times the original speed.

[1]This song is the French version of "Bare Necessities" from the film *The Jungle Book*. Although the original song was in English, the French lyrics do not directly translate the English in order to retain a correspondence with both the song's melody and the intonative features of French.

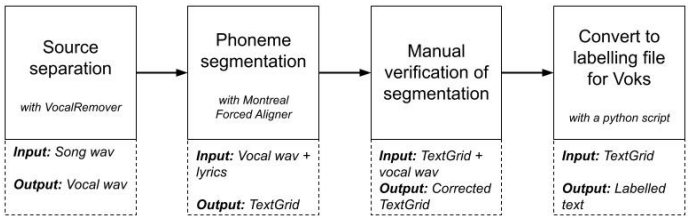
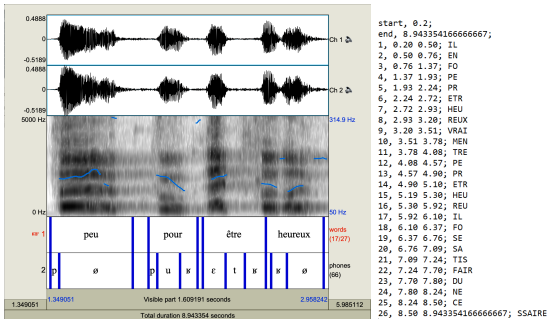


Fig. 3. Data preparation pipeline




```
start, 0.2;
end, 8.943354166666667;
1, 0.20 0.50; IL
2, 0.50 0.76; EN
3, 0.76 1.37; FO
4, 1.37 1.93; PE
5, 1.93 2.24; PR
6, 2.24 2.72; ETR
7, 2.72 2.93; HEU
8, 2.93 3.20; REUX
9, 3.20 3.51; VRAI
10, 3.51 3.78; MEN
11, 3.78 4.08; TRE
12, 4.08 4.57; PE
13, 4.57 4.90; PR
14, 4.90 5.10; ETR
15, 5.19 5.30; HEU
16, 5.30 5.92; REU
17, 5.92 6.10; IL
18, 6.10 6.37; FO
19, 6.37 6.76; SE
20, 6.76 7.09; SA
21, 7.09 7.24; TIS
22, 7.24 7.70; FAIR
23, 7.70 7.80; DU
24, 7.80 8.24; NE
25, 8.24 8.50; CE
26, 8.50 8.943354166666667; SSAIRE
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Fig. 4. (Left) Syllable segmentation generated by Montréal Forced Aligner [13] opened in Praat [3] for manual correction. (Right) Labeling file generated by the python script. Letters to be highlighted in the Unity interface are manually added.

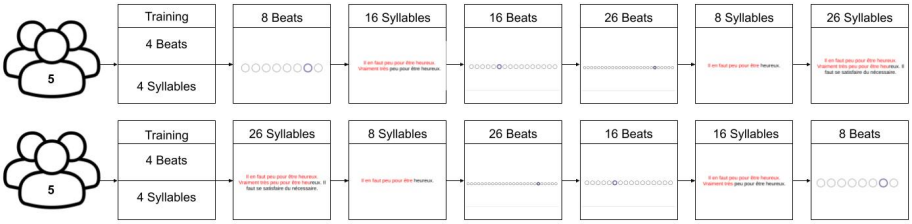

Fig. 5. Diagram of the pilot study showing each stage of the experiment. Subjects were divided into two groups, with one group performing the stimuli in one order and the other the inverse order. Training stimuli had two repetitions each while all other stimuli had 10 repetitions.

To address question two, three sequence lengths were tested, corresponding to the song segment's phrase 1 (8 syllables), phrases 1-2 (16 syllables), and the entire segment (26 syllables). Reference recordings for the three lengths were generated for both conditions.

4.2 Task

The study took place on a laptop computer in a quiet room at a research lab, and sound was heard through headphones. Figure 5 depicts the protocol, which included 6 stimuli (8-syllable speech, 8 syllable beep, 16 syllable speech, 16 syllable beep, 26 syllable speech, 26 syllable beep). For each stimulus, subjects had 10 trials. Each trial consisted of listening to the reference for the target stimuli and then trying to reproduce it with audio feedback from Voks and visual feedback on the screen. The reference recording could only be heard once during each trial, but subjects triggered the playback themselves by pressing a key on the keyboard. Subjects could also decide when to begin their imitation of the reference. Timing data was recorded with the first press for each trial as the zero-point. When all 10 trials are completed for one stimuli, the change in the visual feedback signaled the start of a new stimuli. Subjects were instructed to not stop in the middle of an imitation round but were free to rest between trials.

Subjects began the study started with a familiarization task. 4-syllable stimuli for both conditions was used, with two repetitions each. To address question 3, subjects completed a short questionnaire about their perception of the speech versus beep conditions after the study (see Appendix A).

5 RESULTS

The average error for each trial is computed as the mean of the absolute deviations between the intervals of the trial and the target rhythm, where the intervals are defined the durations between two consecutive events (syllables or beeps).

If we note X_1,...X_n the timestamps in the target stimulus and Y_1,...Y_n the timestamps in a trial, where n is 8, 16, and 26 for stimuli of different lengths, the average error, $\overline{\epsilon}$, is expressed by equation 1.

$$\overline{\epsilon} = \frac{\sum_{i=0}^{n-1} |(Y_{i+1} - Y_i) - (X_{i+1} - X_i)|}{n} \tag{1}$$

Table 1 shows descriptive statistics of average error for each stimulus. Median average error of 16-event conditions were about twice as high for the 8-event condition. The 26-event condition showed median values about 4 times higher than for the 8-event condition, indicating an effect of sequence length on the difficulty of the imitation task. This effect was found to be significant for both syllable and beep stimuli (ANOVA χ^2 for 8 vs 16 vs 26 beep: (N=10, df=2)=13.4000, p=0.00123, for 8 vs 16 vs 26 syllables: (N = 10, df = 2) = 7.2000 p = .02732).

Wilcoxon matched pairs tests failed to show significant differences between beeps and syllables phrases of the same length. Friedman's ANOVA failed to find an effect of repetition on subject's performance. These results could be due to the small sample size of subjects and large variability in mean errors.

Given the normal distribution of motivational inventory survey data we used a repeated measures ANOVA to analyze the results. This analysis showed a significant interaction between Stimuli (Beep/Speech)*Cat (inventory category) (with F=12,479, P<0,001). Tukey posthoc tests, corrected for multiple comparisons, showed a significant higher rating for Hard in Beep vs Speech and Satisfied in Speech vs Beep (P<0.05). In other words, the beep task was perceived to be more difficult than the speech task and subjects were more satisfied with their performance in the speech versus the beep condition. Results also showed a tendency for higher ratings in the Fun and Interesting categories for speech. These differences were not significant in our small sample and need to be explored in a larger sample.

Table 1. Descriptive statistics for the average error ($\overline{\epsilon}$) in seconds of the trials of different stimuli.

	Mean	**Median**	**Min**	**Max**	**Std Dev**
8 beeps	0.1570	0.1054	0.0744	0.4884	0.1274
8 syllables	0.1595	0.0891	0.0784	0.4611	0.1283
16 beeps	0.2737	0.2625	0.0895	0.6376	0.1592
16 syllables	0.2800	0.2116	0.0849	0.5980	0.1747
26 beeps	0.7178	0.3900	0.2051	1.9470	0.6555
26 syllables	0.6144	0.5277	0.2417	1.4379	0.3536

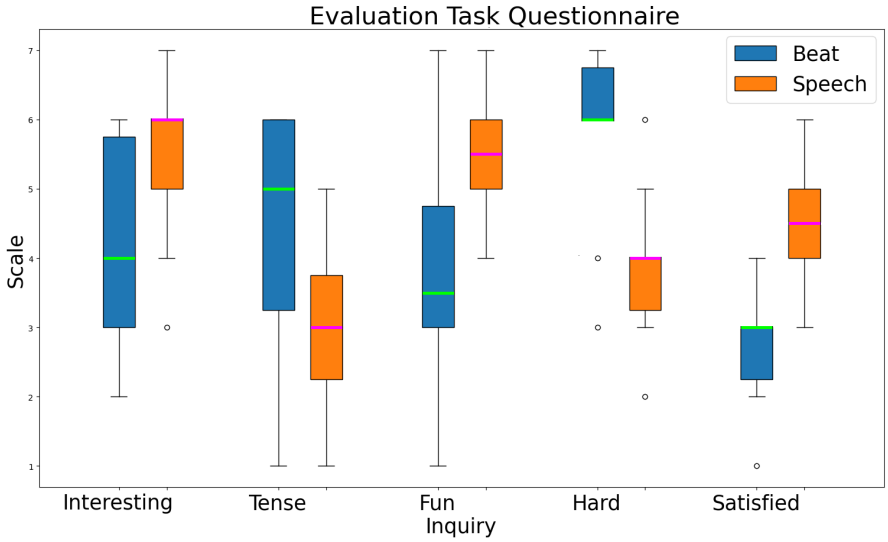

Fig. 6. Boxplot of survey subject responses to the survey at the end of the pilot study (see Appendix A. Responses were given on a 7-point Likert scale (1: not true at all, 3: somewhat true, 7: very true). Responses for the beep condition are shown in blue. Responses for the speech condition are in orange.

6 DISCUSSION & FUTURE WORK

The fact that subjects rated the pilot study task as less difficult for the speech over the beep condition despite no significant difference being found in the performances across conditions is a promising result for the use of Rhythm Karaoke for dexterity rehabilitation. In a rehabilitation context, higher motivation to do an exercise can help patients spend more time practicing, which leads to improvements in performance [27].

Differences in accuracy between stimuli of different syllable lengths suggests that sequence length is an important factor to consider, especially when adapting Rhythm Karaoke for use with neurological patients (e.g., stroke patients). Patients may need shorter lengths than healthy subjects since finger movements are impaired and performed with reduced speed. We can also imagine an application that trains an entire song first in very short phrases, and then with progressively longer phrases as the subject gains more familiarity, with the final goal to tap the entire song.

Even though our pilot study did not find significant differences between the speech and beep imitation tasks, it does not mean that no differences exist. Repeating the experiment with a larger sample size of subjects may uncover significant results.

Additionally, our data for longer phrases and especially for the beep condition had very high variance due to large outliers in the average error. Some subjects did report difficulties in concentration during the 26-beat/syllable conditions. The lower variance found in the 26-syllable versus 26-beep stimuli suggests that the song lyrics helped subjects focus their attention. An interesting question for a future study is whether melody combined with speech help focus subjects attention more than melody alone.

Finally, given the musical nature of our rhythm imitation task, subjects with musical training are expected to exhibit more accurate performance [2]. While we collected data on subjects' musical training, our sample size is too small to claim any significant differences.

7 CONCLUSIONS

This paper presented Rhythm Karaoke, a tapping game to control the syllabic rhythm of real-time vocal synthesis, designed for the rehabilitation of manual dexterity and compatible with the Dextrain Manipulandum. A semi-automatic pipeline was devised that allows the vocal solo line of any song to be extracted and used in our system.

A preliminary evaluation of Rhythm Karaoke was conducted where 10 healthy subjects imitated rhythmic sequences of different lengths, represented by a segment of a familiar song or beeps. Though we hypothesized higher accuracy for the speech condition, our pilot study results did not show a significant difference in performance between the beep/speech conditions. Due to large variance observed in the data, the evaluation should be repeated with more subjects.

However, we did find a significant difference in subjects' self-reported ratings of the two conditions. Subjects indicated significantly higher satisfaction with their speech performance and rated the beep task as more difficult, a promising result toward our overarching goal of creating motivating exercises for dexterity rehabilitation.

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A SURVEY QUESTIONS

Below are the list of statements in the survey for which subjects selected a response on a 7-point Likert scale (1: not true at all, 3: somewhat true, 7: very true). Subjects gave two sets of answers for the statements, for the beep and speech

conditions, respectively. The bold text indicates the name of the category in our analysis and were not shown to the subjects. All questions except #4 came from the Intrinsic Motivation Inventory (IMI) [12].

(1) **Interesting:** I found the task very interesting.
(2) **Tense:** I felt tense while doing the task.
(3) **Fun:** Doing the task was fun.
(4) **Hard:** Doing the task was arduous.
(5) **Satisfied:** I am satisfied with my performance on this task.

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